

Cerebrospinal fluid and the blood-brain barrier

The written notes for the cerebrospinal fluid and the blood-brain barrier walkthrough. Week 5 competency 6.

Cerebrospinal fluid

- Fills the ventricles inside the brain and the subarachnoid space around the brain and spinal cord.
 - Functions: buoyancy (the brain presses on the skull with only a small fraction of its weight), cushioning against sudden movement, and a protected chemical environment.
1. Secreted into the ventricles by the choroid plexus, capillaries covered by specialized ependymal cells.
 2. Flows through the ventricles and out into the subarachnoid space.
 3. Reabsorbed into venous blood at the arachnoid villi, which project into the venous sinuses.
- About 150 mL is present at any time; about 500 mL is made each day, so the volume turns over a few times a day.
 - Blocked outflow: production continues, fluid accumulates, pressure rises within hours. This is hydrocephalus.

The blood-brain barrier

- Brain capillary endothelial cells are joined by tight junctions that seal the gaps found in most capillaries.
- Astrocyte foot processes press on the capillaries and help maintain the tight junctions.

Substance	Crosses?	How
Oxygen, carbon dioxide	Yes	Small and nonpolar: diffuse through the membranes
Lipid-soluble drugs (anesthetics, alcohol, nicotine)	Yes, quickly	Dissolve through the endothelial membranes
Glucose	Yes	Carried by GLUT1 transporters, facilitated diffusion
Amino acids, L-dopa	Yes	Carried by amino acid transporters
Large proteins, water-soluble substances with no carrier (dopamine)	No	No gaps to pass through and no carrier

- Parkinson's disease: dopamine cannot cross, so its precursor L-dopa is given; it crosses on an amino acid carrier and is converted to dopamine in the brain.
- Circumventricular organs lack the barrier so neurons can sample the blood: the area postrema detects toxins and triggers vomiting, and parts of the hypothalamus sense blood osmolarity and release hormones into the blood.